PeTTa Abstract Algorithmic Chemistry

Implementation, Experiments, Results, and Next-Step Questions

External-review packet

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Abstract

This report describes the PeTTa-Chem project from its first executable chemistry kernel through its most recent autocatalytic-set experiments. The project has built a substantial PeTTa-native symbolic reaction substrate with deterministic replay, bounded candidate generation, explicit provenance, event histories, first-class catalysis relations, conservative cycle scanners, RAF detectors, controlled guidance policies, causal ablations, and preregistered experiment matrices. The central scientific result so far is negative but informative: the system reliably recovers planted and deliberately favorable RAF structures, but no completed protocol supports spontaneous autocatalytic-set emergence. One frozen guided condition produced a strong bounded uplift, but independent replication failed and a larger successor made weak guidance worse than the unguided control. This packet is designed to let an external adviser audit the engineering and experimental logic, identify the most valuable next scientific question, and discourage further incremental infrastructure growth without a sharper objective.

Contents

1 Executive assessment

1.1 What exists now

The repository contains a PeTTa/MeTTa chemistry kernel and experiment suite rather than a Python simulation with a symbolic facade. Chemistry state, reaction applicability, candidate generation, bounded selection, checked firing, chamber evolution, event recording, replay, detector inputs, and the large exp07 trajectory matrices execute in PeTTa on the SWI-Prolog-backed PeTTa runtime. Python is restricted mainly to file serialization and two exp04 reference/reduction harnesses.

At the audited head `bfa0e47c2af4bb46dae830a6e7493de97de4051c`, the repository has 308 commits and approximately 42,264 lines across PeTTa source, smoke fixtures, and harness scripts. The active branch is `agent/exp07-stochastic-doob-pilot`. Three untracked items were present and were not treated as authoritative evidence: the draft `exp08/` directory, `experiments/scratch_cat.metta`, and `src/chem_catalysis.metta`.

1.2 What the evidence currently says

- Planted pair and cycle detectors work on positive and negative controls.
- Forty generated-unplanted eight-rule seeds, each evaluated under random, shuffled-catalyst, and no-catalysis families, produced no detected cycles of lengths 2 through 7.
- A deliberately success-biased 20-rule chemistry contains a genuine RAF under the implemented detector, but the result is highly sensitive to catalyst routing and pool design and is not an unbiased emergence result.
- A frozen ordered-bootstrap experiment produced RAF incidence 0/8 unguided, 6/8 weak guidance, and 0/8 shuffled, with all prespecified ablations collapsing the weak result. This was evidence for bounded guided causal uplift in that exact constructed chemistry.
- The independent primary replication failed: 1/16 unguided, 4/16 weak, 0/16 shuffled, below the registered effect thresholds.
- A robustness cohort separated incidence but failed causal attribution: 8/16 unguided, 16/16 weak, 0/16 shuffled, while guidance removal retained 8/16.
- The larger dual-bootstrap successor was negative: 13/32 unguided, 9/32 weak, 0/32 shuffled. Its prospective mechanism audit labeled the failure *applicability loss*: weak guidance made 51 blocked first-absent selections versus 38 for unguided.

1.3 Bottom line

Observed: The project has a credible experimental substrate and has generated well-controlled negative results. It has not demonstrated spontaneous ACS/RAF emergence. The best positive guided result did not independently replicate.

Interpretation: The main bottleneck is no longer absence of replay, detector checks, or candidate-cap provenance. It is experimental identifiability: the current chemistries confound catalytic organization with source ordering, resource availability, applicability, and hand-designed pathway structure.

Boundary: No result in this report supports biological realism, open-ended evolution, semantic chemistry, AGI, or spontaneous autocatalytic emergence.

2 Project objective and epistemic vocabulary

The project asks whether a bounded symbolic rewrite soup can produce autocatalytic organization that is:

1. not explicitly planted in the rule pool;
2. detected by a validated ACS/RAF-like criterion;
3. persistent across time rather than present only as a static graph;
4. reproducible across seeds;
5. discriminated from shuffled and no-catalysis controls; and
6. intervention-relevant, so deleting catalytic structure or key rules destroys the measured effect.

The project deliberately distinguishes several notions:

Static catalytic cycle
A graph-level product/catalyst cycle. It may be unreachable or dynamically irrelevant.

RAF
A reflexively autocatalytic, food-generated reaction subset under the implemented closure test.

Active RAF/ACS
A qualifying set that occurs in a realized event window.

Causal RAF/ACS
An active set whose prespecified structural ablation removes the endpoint.

Guided uplift
A difference caused by a fixed selection policy in a constructed chemistry; this is not spontaneous emergence.

Spontaneous emergence
A stronger claim requiring non-planted chemistry, appropriate controls, replication, and causal discrimination. Its status is currently `none`.

3 Implemented system

3.1 Runtime and module organization

Version 0.1 targets the local PeTTa runtime backed by SWI-Prolog 9.3.x (initial validation used 9.3.36). The main source modules are:

Module	Implemented responsibility
`chem_dynamics.metta`	Generic symbolic chemistry state and dynamics support.
`chem_exp00.metta`	Molecule/rule/state/chamber schemas, candidate generation, caps, deterministic selection, applicability, checked firing, event history, replay, hashes, bounded tick loops, attrition provenance, and stop dispositions.
`chem_exp01.metta`	Planted ACS fixtures, pair scanner, catalytic cycle scanners through length seven, and replay-derived productivity ablation.
`run_contract.metta`	Consequential-run schema for configuration, manifest, events, snapshots, metrics, ACS candidates, ablations, summaries, and replay status.
`chem_exp02.metta`	Seeded generated rule families, random/shuffled/ no-catalysis controls, folded summaries, and cycle-scan reports.
`chem_exp04.metta`	Rich ligation/cleavage/modification chemistry, first-class catalysis facts, seeded positive RAF fixtures, and PeTTa report atoms.
`chem_exp05.metta`	Soft affinity-biased catalyst assignment and rotated controls.
`chem_exp06.metta`	Favorable-condition/bridge probes, bounded intervention tables, stateful candidate-cap paths, and ACS boundary checks.

Module	Implemented responsibility
`chem_exp07.metta`	Weighted categorical policies, stateful ensembles, rolling RAF detectors, matched controls, ordered and dual bootstrap chemistries, causal ablations, exact replay, and mechanism diagnostics.

3.2 Core data and transition contracts

The core state is explicit and inspectable. Molecules carry symbolic identity and integer abundance. Rules carry stable ID, reactants, product, and catalyst. A chamber contains state, source rules, and event history. A candidate records the proposed executable reaction, while a candidate pool and cap record which bounded list was exposed to selection.

The checked tick path is conceptually:

source chamber → bounded candidate generation → attrition/provenance validation → deterministic or weighted selection → source-rule identity and applicability checks → firing or explicit no-op → new chamber and event history → exact replay check

Important invariants include nonnegative abundances, exact source-rule ownership, stable candidate identity, fail-closed behavior for stale or foreign candidates, one evaluated generation path per tick, explicit dropped-candidate counts, and equality between recorded and replayed terminal chambers.

3.3 Bounded generation and execution

The exp00 driver has been extended from a one-reaction smoke to:

- candidate caps from zero through eight, including nonpositive and oversized-cap behavior;
- source-rule ownership and generation-count audits;
- explicit requested/effective cap, retained count, dropped count, remaining capacity, attrition stage, and validity dispositions;
- live histories of up to nine eight-tick sweeps and 64 retained events;
- early `stop-quiescent` and explicit `stop-bounded`;
- exact final-chamber preservation and run-wide productive/no-op/ unaccounted outcome totals; and
- total bounded generation for the current 31-rule/four-molecule fixture, while preserving the stable first-eight generated prefix.

The last point is engineering coverage, not a larger scientific sweep. At the current head, 31 source rules are auditable, eight are retained by the fixed execution cap, and 23 omissions are reported explicitly.

3.4 Detection and causal analysis

Three detector families have been implemented:

1. conservative exact product-as-catalyst cycle scanners for lengths 2–7;
2. maximal-RAF pruning with food closure and greedy core minimization, initially mirrored by a host reference and later fed by first-class PeTTa `catalyzes` facts; and
3. event-window detectors for realized three- and four-rule RAF pathways in exp07, coupled to replay and structural ablations.

Detector validation precedes scientific use. Planted cycles are recovered, distractors and no-catalysis fixtures are rejected, the exp04 seeded positive returns maximal RAF 15/core 2 and empty-catalysis RAF 0, and repeated detector calls are invariant to intervening calls and path metadata.

3.5 Guidance and intervention machinery

The later experiments compare:

- unguided categorical selection;
- weak Doob-h or frontier guidance;
- shuffled guidance at matched shifted-mass cost;
- guiding-term removal;
- bootstrap deletion;
- full catalyst-set deletion; and
- individual pathway-rule deletion.

These are bounded causal diagnostics on constructed chemistries. They do not implement an actual Schrödinger bridge solver over learned transition kernels; "Doob-h" denotes a frozen weak reweighting policy inspired by that formalism.

4 Experiment history and outcomes

4.1 Summary table

Exp.	Question	Main result	Status / limitation
00	Does the PeTTa chemistry kernel execute and replay exactly?	Schemas, bounded generation/selection/firing, 64-event histories, provenance, and stop rules implemented.	Engineering validation; no emergence endpoint.
01	Can known ACS structures be recovered and ablated?	Planted two-rule closure, cycles of length 3–7, distractor rejection, and replay-derived productivity drop pass.	Positive detector calibration only.
02	Do simple generated-unplanted pools contain cycles?	40 seeds × 3 families = 120 records; zero cycles for every size 2–7. Planted controls are detected.	Negative for the tested generator and strict cycle criterion.
03	Can generated pools drive multi-tick stateful dynamics?	Productive bridges, controls, candidate caps, event accumulation, replay, and artifact exports implemented.	Mostly constructed productive fixtures; a current-head singleton regression is described in Section 5.
04	Can a richer favorable chemistry sustain a RAF?	20-rule favorable pool: maximal RAF 15, core 2, no-catalysis RAF 0; 56-event dynamics.	Positive but deliberately biased; shuffle sensitivity is large.
05	Can soft catalyst affinity replace hard assignment?	Affinity and rotated-control pools feed ordinary tick machinery.	Plumbing and calibration only; no RAF claim.
06	Can favorable-condition paths bridge ACS-negative to RAF-rich states?	Cap changes alter selected rules and paths, but tested boundary pairs are not ACS-positive.	Diagnostic path scaffold, not an optimized bridge or emergence result.
07	Does weak guidance causally increase realized RAF closure?	One ordered-bootstrap cohort passed; independent replication and dual-bootstrap successor failed.	Program closed as negative for general guided uplift; spontaneous emergence remains none.
08	Is closure robust across every source-order permutation against a degree-preserving catalysis null?	Draft only; no registered run.	Unfrozen/unexecuted with unresolved executable and confidence-bound gates.

4.2 exp00–exp03: substrate and simple generated chemistry

exp00. The first smoke implemented `A+B -Cat-> AB`; later commits generalized it into the common chamber kernel. The current body of work is strong on deterministic replay and fail-closed provenance. It is weak as scientific evidence because most fixtures are deliberately constructed to exercise a code path.

exp01. The planted reciprocal pair is recovered beside distractors, while a non-catalytic molecule cycle is rejected. Removing one planted rule reduces replayed productivity by one event. Dedicated fixtures validate exactly one cycle at each size 3–7.

exp02. The systematic deterministic sweep comprises four planted reciprocal-pair points and 40 generated-unplanted points. Each point has random-polymer, shuffled-catalyst, and no-catalysis records.

Family	Points	Records	Active records	Active pairs
Planted reciprocal control	4	12	4	7
Generated-unplanted	40	120	0	0

The generated-unplanted records also contain zero active cycles for every tested size 3–7. Thus the null is not "the detector never fires": the planted fixtures fire, while the tested generator does not.

exp03. Stateful dynamics were built over the same source families, including productive cap-3 through cap-8 bridges, multi-tick accumulation, shuffled and no-catalysis controls, run exports, and artifact writers. These fixtures demonstrate mechanics and replay but do not establish non-planted ACS formation.

4.3 exp04: favorable rich RAF and bias-reduction sweep

The rich chemistry uses 20 hand-designed ligation, cleavage, and modification rules over food molecules A–D. A specificity-filtered structural catalyst map, food replenishment, and a slow basal trickle make it deliberately favorable.

Observed: The strict host reference found maximal RAF size 15 and greedy core size 2 (`lCD`, `lBCD2`); the core passes the RAF predicate. Empty catalysis gives RAF size 0. A 56-tick run yields 56 events and maintains the food floor.

The reduction sweep changed catalysis mode, basal interval, and pool construction. Representative rows are:

Pool	Catalysis	RAF	Core	Events	Catalyzed
Hand-designed	specific, basal-4	15	2	56	42
Hand-designed	shuffled, basal-4	10	3	53	39
Hand-designed	none, basal-4	0	0	14	0
Generated template	specific, basal-4	0	0	14	0
Generated template	shuffled, basal-4	1	1	56	42

Interpretation: The detector can identify a real RAF in the implemented sense, but the positive structure is largely a property of the engineered pool/catalysis map. Several shuffles retain RAF-like structure, while the mechanically generated pool under specific catalysis is negative. This is an appropriate positive control and a warning against calling static RAF detection "emergence."

4.4 exp05–exp06: affinity and bridge diagnostics

Exp05 assigns catalysts through token-overlap affinity with a positive baseline and rotated-weight control, then projects the result into ordinary candidate pools. Exp06 defines a small intervention space spanning affinity, basal replenishment, cap, and catalysis offset. It includes a bounded shortest-path table and stateful cap probes.

Observed: In the stateful cap probe, cap 1 repeats `e5r0`; cap 2 exposes `e5r1`; both later no-op after reactant A is exhausted. Neither selected two-rule path passes the product/catalyst closure boundary.

Interpretation: Candidate access changes trajectories, but the current bridge scaffold does not yet establish a path into an ACS-positive region. It also does not estimate an uncontrolled transition kernel, solve a Schrödinger bridge, or separate guidance from ordinary resource/applicability effects.

4.5 exp07: guided closure program

4.5.1 N=20 two-rule matrix

Seeds 7–10, ticks 3–22, cap 2, and replenishment at ticks 8, 13, and 18 produced eight events per seed. RAF incidence was 4/4 in all three arms. This was correctly interpreted as an all-arms-positive pool: the chemistry closed independently of guidance.

4.5.2 Rich shared-pool ensemble

For seeds 11–18, rolling three-rule RAF incidence was 5/8 unguided, 7/8 weak, and 3/8 shuffled. Persistence totals were 13/49/11. Both controls remained positive, and guiding removal did not isolate a guidance-caused effect.

4.5.3 Ordered bootstrap

The successor removed the initially available cycle catalyst and forced the path `ob0 -> op1 -> op2 -> op0`. On held-out seeds 19–26:

Arm	RAF incidence	Persistence total	Distinct-rule total
Unguided	0/8	0	42/64
Weak Doob-h	6/8	41	59/64
Shuffled	0/8	0	39/64

All 24 trajectories replayed. Removing guidance, the bootstrap, all cycle catalysts, or any one cycle rule reduced weak incidence to zero. Under the frozen rule, this supported bounded guided causal RAF uplift in that exact chemistry.

4.5.4 Independent ordered replication

The exact-form cohort A used fresh seeds 101–116:

Arm	Incidence	Persistence	Productive events
Unguided	1/16	1	259
Weak	4/16	15	263
Shuffled	0/16	0	250

The weak advantage was below both preregistered six-seed thresholds, so the primary replication failed. Structural ablations were all null, but that did not rescue the failed primary comparison.

Cohort B changed order and draw hash. Incidence was 8/16 unguided, 16/16 weak, and 0/16 shuffled. The incidence separation passed, but removing guidance retained unguided 8/16, above the registered causal maximum of two. Order/hash robustness therefore failed.

4.5.5 Dual-bootstrap successor

The larger successor fixed a six-step bootstrap/four-rule RAF, twelve-rule source, cap 8, seeds 201–232, and ticks 3–34.

Arm	Incidence	Persistence	Events	Diversity
Unguided	13/32	95	549	336
Weak	9/32	51	523	278
Shuffled	0/32	0	396	219

All 96 trajectories replayed. Removing either bootstrap, all four cycle catalysts, or any cycle rule reduced weak incidence to 0/32. Removing guidance retained unguided 13/32. The frozen guided-uplift criterion failed.

The post-result mechanism audit was prospectively specified before deriving new diagnostics. It reproduced registered endpoints and applied a fixed precedence rule. Weak guidance produced 51 blocked selections of the currently first-absent pathway rule, versus 38 for unguided, yielding the label *applicability loss*. This explains the negative result descriptively: guidance often favored a frontier reaction before its reactants were available. It is not a new causal treatment result.

4.6 exp08 draft

The current untracked exp08 draft proposes all 24 permutations of four RAF rules, each paired with a degree-preserving catalyst-routing null, plus exact McNemar inference and mechanistic deletion. Its scientific intent is good: separate catalyst routing from catalytic degree and source order.

It is not ready to run. The draft explicitly leaves unresolved:

- the bit-exact run-0 reference trace and hash;
- executable permutation and null constructors with invariance fixtures;
- shared paired random-stream/cap implementation;
- serialized detector artifact;
- a published lower confidence-bound method for paired incidence difference; and
- deletion-replay implementation and fixtures.

No exp08 result exists, and the draft must not be cited as a preregistered experiment.

5 Current reproducibility audit

On 26 July 2026 the canonical scripts were rerun locally at head `bfa0e47`. Counts below are terminal check marks printed by the PeTTa test harness.

Script	Exit	Passed checks	Failed checks
`run_exp00.sh`	0	513	0
`run_exp01.sh`	0	68	0
`run_contract.sh`	0	35	0
`run_exp02.sh`	0	517	0
`run_exp03.sh`	1	38	1
`run_exp04.sh`	0	79	0
`run_exp04_first_class_catalysis.sh`	0	79	0
`run_exp05.sh`	0	94	0
`run_exp06.sh`	0	61	0
`run_exp07.sh`	0	309	0

The exp03 failure is a current-head reproducibility defect. The query `exp03-exp02-after-3 random seed-11` returns eight identical chamber answers where the smoke expects one. The chamber value itself matches, but duplicate logical answers violate the singleton contract. The script exits 1. This must be fixed and covered before declaring the entire current suite green. It does not alter the stored exp07 endpoint values, which are pinned and independently replay-tested by exp07, but it weakens the claim that every historical layer composes cleanly at current head.

6 What is strong, what is weak, and what is missing

6.1 Strong aspects

- **PeTTa-first implementation.** Most scientific logic is in inspectable symbolic clauses.
- **Replay discipline.** Large matrices require exact trajectory replay and preserve explicit event histories.
- **Negative controls.** Shuffled, no-catalysis, guiding-removal, and structural deletion controls are routine.
- **Detector calibration.** Planted positives and negative fixtures exist before detector use.
- **Preregistration discipline.** Later protocols freeze seeds, horizons, pools, endpoints, costs, and stopping rules.
- **Honest negative conclusions.** Failed replication and causal gates were not rescued by post hoc reinterpretation.

6.2 Weak aspects

- **Excess infrastructure accretion.** Hundreds of small commits expanded cap and provenance coverage far beyond what was needed for the next scientific discrimination.
- **Constructed chemistry dependence.** Most positive RAFs depend on hand-designed pathways, replenishment, basal events, or guidance.
- **Applicability-blind guidance.** Frontier weighting can increase mass on reactions that cannot yet fire.
- **Static/dynamic ambiguity.** A pool can contain a RAF while realized dynamics never reaches or maintains it; some reports mix these levels.
- **Limited statistical generalization.** Several matrices use deterministic hash draws rather than a clearly defined stochastic process with estimable sampling uncertainty.
- **Current regression.** exp03 violates a singleton answer invariant at current head.

6.3 Missing scientific components

- A generative distribution over reaction rules and catalysis maps that is defined independently of desired RAF pathways.
- A scalable, generic RAF implementation validated against a standard reference library or exhaustive small-system oracle.
- Separation of structural RAF existence, dynamical reachability, persistence, and causal productivity into distinct endpoints.
- Replicated parameter sweeps with uncertainty, rather than additional hand-selected pathway designs.
- A neutral resource model: influx, dilution/decay, reaction propensities, and population-size scaling.
- A principled baseline from random catalytic network/RAF theory.

7 Recommended next plan

7.1 Recommendation 0: stop feature accretion and repair reproducibility

Before new scientific work:

1. fix the exp03 duplicate-answer regression and add an explicit uniqueness test at the underlying clause boundary;
2. run the complete canonical suite with compact logs and archive a machine-readable test manifest;
3. commit or archive the untracked exp08 draft and scratch files so the scientific source of truth is unambiguous; and
4. tag a review baseline commit.

7.2 Recommendation 1: define one neutral random-chemistry model

Replace the next pathway-specific chemistry with a small published-style random catalytic reaction-system model:

- polymers over a fixed alphabet with ligation/cleavage;
- a declared food set and maximum polymer length;
- catalysis edges sampled independently at rate p or from one explicitly declared structural kernel;
- stochastic mass-action or Gillespie-like reaction events;
- controlled influx and dilution/decay; and
- no pathway-aware selection or terminal target.

The first question should be calibration, not novelty: does the implementation recover the expected finite-size RAF transition as catalysis density varies? This gives an external theoretical anchor and tells us whether the detector and dynamics behave plausibly.

7.3 Recommendation 2: validate the RAF detector against an oracle

For small rule sets, enumerate every subset and compare the PeTTa maximal-RAF and core outputs with an independent reference implementation. Include:

- food-generation closure;
- catalyst-in-closure checks;
- multiple catalysts and multiple products where supported;
- degenerate self-catalysis;
- unreachable static cycles;
- monotonicity when catalysis edges are added; and
- deletion sensitivity.

This is the highest-value application of the project's "validate the miner first" research rule.

7.4 Recommendation 3: separate four endpoints

Every future run should report, separately:

1. **structural**: does the sampled reaction system contain a RAF?
2. **reachable**: can its molecules be generated from food under the actual dynamics?
3. **active/persistent**: does it fire and persist for a specified fraction of a post-burn-in window?
4. **causal/productive**: does deleting its catalysis reduce a declared productivity or persistence endpoint against a matched null?

Passing a weaker endpoint must never be described with the stronger label.

7.5 Recommendation 4: only then revisit guidance

If guidance remains scientifically interesting, make it applicability-aware: weight only currently executable reactions, or explicitly factor guidance into "reach prerequisite" and "close RAF" potentials. Compare against:

- unguided stochastic dynamics;
- an applicability-only heuristic with no RAF knowledge;
- degree-preserving shuffled guidance;
- matched compute/intervention cost; and
- a true bridge/control solution on a tiny state space where the optimal policy can be computed exactly.

The exact small-state solution is essential before using "Doob transform" or "Schrödinger bridge" as more than inspiration.

7.6 Recommendation 5: postpone exp08 in its current form

The 24-permutation paired catalyst-routing test is careful, but it studies the same constructed dual-bootstrap pathway whose general guided effect has already failed. Running it now risks answering an increasingly narrow question about one artifact. Complete it only if external advisers believe catalyst-routing identifiability in that exact model is intrinsically valuable. Otherwise, reuse its good design ideas—paired degree-preserving nulls, exact replay, mechanistic deletion, and valid paired inference—inside the neutral random chemistry calibration.

8 Questions for external advisers

1. Is the recommended neutral random catalytic reaction-system model the right next substrate, or is there a stronger standard framework/library we should adopt directly?
2. Which established RAF algorithm and benchmark suite should be used as the independent oracle?
3. What is the cleanest finite-size phase-transition calibration for this scale: catalysis probability, polymer length, food-set size, or another control parameter?
4. Should the scientific target be structural RAF formation, dynamically active RAF persistence, or ecological selection among RAFs?
5. Is an exact Schrödinger bridge/Doob-control baseline valuable here, or does it distract from unbiased emergence?
6. Are degree-preserving catalyst shuffles sufficient nulls, and what other invariants should be preserved?
7. What minimal stochastic resource model avoids both trivial starvation and hidden terminal forcing?
8. Which part of the present PeTTa implementation should be retained, simplified, or replaced before the next experiment?

9 Reproduction and evidence map

Repository: https://github.com/bgoertzel-sing/petta-chem. Audited local head: `bfa0e47`; the full object is recorded in the project status.

Canonical local commands:

```
scripts/run_exp00.sh
scripts/run_exp01.sh
scripts/run_contract.sh
scripts/run_exp02.sh
scripts/run_exp03.sh
scripts/run_exp04.sh
scripts/run_exp04_first_class_catalysis.sh
scripts/run_exp05.sh
scripts/run_exp06.sh
scripts/run_exp07.sh
```

Primary evidence pointers:

- `README.md`: current implementation and experiment summary.
- `experiments/exp02_small_sweep_20260630/SUMMARY.md`: 40-seed generated-unplanted cycle results.
- `experiments/exp04/RUN.md`: rich RAF design and host refinement.
- `experiments/exp04_reduction_sweep_20260708/SUMMARY.md`: pool/catalysis/basal reduction matrix.
- `experiments/exp07/PREREG_ORDERED_BOOTSTRAP.md`: first guided positive result.
- `experiments/exp07/PREREG_ORDERED_REPLICATION.md`: failed primary replication and failed robustness cohort.
- `experiments/exp07/PREREG_DUAL_BOOTSTRAP.md` and `DUAL_BOOTSTRAP_REPORT.md`: larger negative successor.
- `experiments/exp07/PREREG_DUAL_MECHANISM_AUDIT.md`: prospective diagnostic plan; implementation result is `applicability-loss`.
- project records under `projects/petta-chem/`: chronological status, decisions, tasks, and notes.

10 Conclusion

PeTTa-Chem has succeeded as an inspectable research-engineering substrate. It has not yet succeeded at its central scientific milestone. The accumulated evidence says:

> The implementation can represent, replay, detect, guide, and ablate autocatalytic structures. The tested simple random generators do not produce them; favorable hand-designed systems do; and guidance effects are fragile under independent replication and applicability/resource changes.

The most productive next move is therefore not another cap, seed family, or pathway-specific guidance protocol. It is a reset around a neutral random chemistry with an independently validated RAF oracle and a known finite-size baseline. That would convert the existing engineering investment into a cleaner scientific instrument.